Classification of Abalone Age Using Neural Networks

Abstract

This report presents a neural network model for classifying abalone age groups using their physical measurements. The study followed a structured framework of data preprocessing, feature exploration and systematic experimentation to rigorously tune model hyperparameters such as hidden units, learning rate and optimizer choice. A multi-layer perceptron model was developed and trained using TensorFlow Keras and evaluated using metrics such as accuracy, confusion matrix and ROC-AUC. The model overall achieved a strong performance on younger abalone classes but struggled with older classes due to class imbalance. The results highlight the effect of network depth and exploration of optimisation strategy on model performance.

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1. Introduction

The purpose of this report is to use data processing and neural networks to classify the age of abalones based on their physical measurements. Abalone age is traditionally measured by cutting the shell, staining it, and counting the growth rings through a microscope — a process that is accurate but extremely slow and repetitive. To address this inefficiency, this project applies machine learning techniques to predict abalone age from easily obtainable physical attributes.

The dataset used in this report is sourced from the UCI Machine Learning Repository and contains 4,177 abalone samples, each described by eight attributes including *Sex, Length, Diameter, Height,* and various weight-based measurements . The true age in years is estimated as the number of rings plus 1.5 years. For this classification task, age values were divided into four groups: ≤ 7 years, 7 – 10 years, 10 – 15 years and > 15 years (Appendix 1), each named from class 1 to 4 respectively.

The first part of this report focuses on data processing and exploratory analysis. This includes cleaning the dataset, one-hot encoding the categorical *Sex* variable, examining feature correlations and visualising feature distributions using histograms and boxplots. The processed data is then split into training and testing sets (60/40) to prepare it for modelling.

The second part applies dense neural network models to classify abalone ages into the four defined groups. The experiments are structured to investigate key architectural and optimisation parameters including the number of hidden neurons, learning rate, number of layers and choice of optimiser (SGD vs Adam). Model performance is evaluated using accuracy across ten experimental runs with our final model using confusion matrices and ROC-AUC curves as well.

The findings are discussed in terms of how neural network design decisions influence classification accuracy and generalisation ability across abalone age groups.

2. Data Selection and Preparation

The abalone dataset used in this study was obtained from the UCI Machine Learning Repository. The dataset contains 4,177 records of individual abalones, each containing 7 physical attributes, one categorical variable (Sex) and its rings. These attributes describe measurable shell and body dimensions that correlate with the biological age of the abalone. Each record includes:

- Sex (categorical): male (M), female (F), or infant (I)
- Length (continuous): longest shell measurement in millimetres

- Diameter (continuous): perpendicular to the length
- Height (continuous): with meat in shell
- Whole weight (continuous): total weight of the abalone in grams
- Shucked weight (continuous): weight of the abalone meat in grams
- Viscera weight (continuous): gut weight after bleeding
- Shell weight (continuous): dry shell weight in grams
- Rings (integer): number of shell rings used to estimate age

Since each ring corresponds approximately to one year, the true biological age of each abalone was estimated as Age = Rings + 1.5 years.

Because the original dataset did not include column headers, descriptive names were manually assigned. The 'Rings' variable was transformed into a new continuous variable 'Ring-Age', before being split into the four age categories following appendix 1.

The categorical feature *Sex* was then **one-hot encoded** into three binary columns (*Sex_M*, *Sex_F*, *Sex_I*) to allow the neural network to interpret non-ordinal categories numerically. This is important as it prevents the model from assuming any artificial ordering or hierarchy between the sex categories, which could be misleading to the model's learning process.

Standardisation was not explicitly required, so all numerical variables were retained on their original scales. Neural networks can adapt to different magnitudes, and our focus was on evaluating architecture and optimiser choices.

Finally, the dataset was divided into training (60%) and testing (40%) subsets using stratified sampling to ensure the same class distribution in both splits. Appendix 2 justifies this correction, as it confirms a significant imbalance with younger abalones dominating the dataset. Appendix 2 is a bar plot showing the distribution of abalones relative to their age with both raw values and percentages. Within the graph, we observe that Class 3 dominates the outcome space, accounting for 54.58% of the dataset, followed by Class 2 at 29.16%, with Classes 1 and 4 contributing just over 16% combined. Even with a stratified sampling approach, this imbalance heavily influenced the classifier's accuracy for older abalones, as discussed in later sections.

3. Exploratory Data Analysis

Following data cleaning and class grouping, exploratory data analysis (EDA) was performed to better understand the distribution and relationships among the features. This process helped identify patterns, outliers, and correlations that could influence neural network performance.

3.1 Feature Distributions

Each continuous variable was visualised using histograms to inspect the distribution of measurements across the dataset (Appendix 3-9). The results indicated that most weight-related attributes — Whole Weight, Shucked Weight, Viscera Weight and Shell Weight — exhibited right-skewed distributions, showing that a large majority of abalones are smaller or lighter, with only a few heavier specimens. On the contrary, Length, Diameter, and Height showed approximately bell-shaped distributions although still being left-skewed but still implying a strong concentration of mid-sized individuals.

Such skewed distributions highlight the physical variation within the dataset and justifies the use of a non-linear model such as neural networks which can capture such complexity.

3.2 Boxplot Analysis

Boxplots were generated for each feature to examine data spread and potential outliers (Appendix 10–16). The Height attribute displayed minimal variation, while the weight-related features had noticeably larger interquartile ranges, reflecting the broader diversity in abalone mass. Several outliers appeared at the upper ends of these weight distributions, most likely representing older abalones with thicker shells and heavier bodies. These visual trends align well with biological expectations, suggesting that as abalones age, their mass and shell density increase, leading them to fall into the higher age classes.

3.3 Feature Correlation

A correlation heatmap (Appendix 17) was used to visualise the linear relationships among all numerical features. The heatmap revealed clear evidence of multicollinearity: Length, Diameter, and Whole Weight were highly correlated with one another (0.93-0.99). Similarly, all weight-related variables — Whole Weight, Shucked Weight, Viscera Weight and Shell Weight — exhibited strong positive correlations above 0.88.

Interestingly, the encoded Sex variable showed negative correlations with all continuous variables however, this does not represent a meaningful linear trend as the relationship arises from the one-hot encoding process rather than an actual numerical relationship.

4. Methodology

This section describes the experimental framework used to construct, optimise and evaluate neural networks for classifying abalone age groups. Models were implemented in Python with TensorFlow/Keras and refined sequentially to ensure a clear rationale for design decisions. We repeated all experiments 10 times and used the same initial weights for corresponding runs across models so that observed differences stemmed from design changes rather than random initialisation. Training is used with best-weight restoration to prevent overfitting. This approach enabled systematic evaluation of performance trade-offs and helped identify configurations that improved generalisation.

All experiments were repeated 10 times to ensure the reliability of results. To ensure that these comparisons are fair and unbiased, the same initial weights were used for corresponding runs across all models. This approach eliminates variation that may come from random weight initialisation ensuring that any observed performative metric differences are caused by the model design rather than random chance.

Model training was conducted for a maximum of 120 epochs with a standard batch size of 64. The batch size provided a balance between computational efficiency and gradient stability since as smaller batches such as true stochastic gradient descent with batch sizes of 1 may introduce more noise, while very large batches tend to converge too slowly or to suboptimal minima.

Early stopping mechanism was implemented with a patience of 15 epochs to prevent overfitting and unnecessary computation given early convergence. Specifically, training was automatically stopped when the model's loss did not improve for 15 consecutive epochs, and the best weights were restored. This allowed the model to maintain optimal performance on unseen data while avoiding overtraining on the training set.

Overall, these design choices collectively ensured that the modelling process remained both robust and computationally efficient along with reproducible results and controlled variability across experiments and models.

4.1 Optimal Number of Hidden Neurons

The first stage aimed at establishing the optimal number of neurons for a single hidden layer while keeping other parameters fixed. A one-hidden-layer network was chosen as the baseline model since it represents the simplest model capable of capturing the relationships.

Four different hidden layer sizes were tested, consisting of 5, 10, 15, and 20 neurons. The baseline optimizer selected was Stochastic Gradient Descent (SGD) with a constant learning rate of 0.01, which is commonly used as a standard reference. Each configuration was trained and evaluated 10 times using identical initial weights for each run to ensure fair and reproducible comparisons.
The performance metrics in accuracy were averaged across the runs for a hidden layer size.

This is accompanied by its standard deviation and 95% confidence intervals, computed using a t-distribution since the number of runs was small.

This process allowed the investigation of whether increasing model capacity improved learning performance without inducing overfitting as reflected by test accuracy. It also allowed for the optimal hidden layer size that achieved strong generalisation to unseen data.

The results showed that model performance improved as the number of neurons increased, but only up to a certain point. Beyond that, the test accuracy began to drop off, suggesting that adding more neurons no longer provided any more meaningful gains and may have led to slight overfitting. The hidden layer size that achieved the highest average test accuracy was 15 neurons, which was therefore selected as the optimal configuration for the next stage of experiments. The results are shown in Appendix 18, where the accuracy denoted as Acc, is the mean accuracy of the experiment for a given specification with its SD and 95% confidence interval in brackets.

4.2 Optimising the Learning Rate

Given the optimal hidden layer size, the next step focused on fine-tuning the learning rate which was a hyperparameter that controls how quickly or slowly the network updates its weights during training. Four different learning rates were tested: [0.1, 0.01, 0.001, 0.0001], using the same one-layer architecture and experimental setup from the earlier step.

The learning rate directly affects the stability and convergence of the model's optimisation process. If it is too high, the model may overshoot the minima and fail to converge, while if the rate is too low, it can result in extremely slow learning or the risk of getting stuck in a local minimum. To ensure a fair comparison across all settings, each configuration was trained 10 times using identical initial weights so that differences in performance would be due only to the learning rate rather than random weight initialisation.

The mean test accuracy and its variability across these runs were then used to evaluate each learning rate as presented in Appendix 18. Among the tested values, a learning rate of 0.1 achieved the highest mean accuracy while maintaining the lowest standard deviation across runs, indicating stable convergence and generalisation. This rate provided an effective balance between learning speed and stability and was therefore selected as the optimal configuration for the remaining experiments.

4.3 Assessing Network Depth

After determining the optimal number of hidden neurons and learning rate, the next stage explored how the network depth by adding a second hidden layer, to evaluate its effect on performance. This step aimed to investigate whether a deeper architecture could improve the network's ability to capture relationships between the abalone's physical attributes and its age class.

Once again to ensure a fair comparison, the number of neurons in each hidden layer was kept identical using the optimal number calculated earlier along with the learning rate of 0.1. The model was trained with the Stochastic Gradient Descent optimiser with the same early stopping parameters. Each configuration was repeated 10 times, reusing the same fixed initial weights across runs to ensure consistent starting conditions.

Performance metrics calculated across runs to assess the model's reliability and generalisation. The results, summarised in Appendix 18, showed that adding a second hidden layer led to only a slight improvement in accuracy, indicating that most of the model's learning capacity had already been achieved with one layer but nevertheless still improved the overall network.

The results suggests that the abalone classification problem while is non-linear it is still not complex enough to benefit substantially from more complex network.

4.4 Comparison of Optimisation Algorithms

The final experimental stage compared two optimisation algorithms — Stochastic Gradient Descent and Adam to determine which provided better convergence and generalisation for the classification problem. Optimisers play a critical role in adjusting network parameters during training, affecting both the learning speed and stability.

Both models used the same network architecture: two hidden layers with 15 neurons each, an optimal learning rate of 0.1, and identical early stopping and batch size settings. Each optimiser was trained using the same 10 initial weight sets generated by fixed random seeds, ensuring that any observed performance difference was due solely to the optimiser's behaviour and not random weight initialisation.

When comparing results, Adam achieved higher training accuracy and produced slightly better test accuracy, showing that it was able to fit the data efficiently without overfitting. Its adaptive learning rate mechanism allowed for smoother and faster convergence, leading to lower variability across the 10 runs. In contrast, SGD exhibited slower learning progress and greater variability between runs, suggesting slower convergence and a greater tendency to get stuck in local minima.

Overall, Adam was selected as the final optimiser for the optimal model due to its superior stability, higher training accuracy, and stronger test performance. The comparison between both optimisers highlights how adaptive learning methods like Adam can significantly improve performance on moderately complex datasets such as the Abalone classification problem.

5. Results and Discussion

The final model evaluation incorporated both quantitative metrics and visual diagnostics to assess classification accuracy, class balance and model reliability. All experiments were repeated ten times to ensure consistency, with mean accuracy and standard deviations reported to account for random variability.

The compiled results (Appendix 18) showed progressive improvements through successive refinements. After optimising the hidden layer size, learning rate and optimiser, the highest overall performance was achieved using a two-layer neural network with 15 neurons per layer, the Adam optimiser and a learning rate of 0.1. This configuration yielded a mean training accuracy of 0.71 ± 0.01 and a test accuracy of 0.72 ± 0.01, indicating both strong generalisation and consistent convergence.

The confusion matrix (Appendix 19) provides insight into the class-level behaviour. The model showed strong predictive performance for mid-aged abalones (Classes 2 and 3), which together constitute more than 80% of the dataset. On the other hand, it struggled with the youngest (Class 1) and oldest (Class 4) abalones, often misclassifying them as neighbouring classes. This pattern is consistent with the class imbalance mentioned earlier, where Class 3 dominated with over half of all samples, whilst Class 1 only accounted for 4.52%. This extreme distribution likely restricted the network's ability to form robust decision boundaries for those groups. Nonetheless, the misclassifications remained within adjacent age categories, which is somewhat reasonable for a continuous biological process like ageing.

The ROC curves illustrated in Appendix 20 further illustrate model discrimination. The overall macro-average AUC reached 0.88, confirming strong class separability. Individual AUCs were 0.98 for Class 1, 0.87 for Class 2, 0.79 for Class 3 and 0.86 for Class 4. Although Class 4 performed poorly in the confusion matrix, its relatively high AUC of 0.86 indicates that the model still produced confident probability estimates but had difficulties with threshold-based classification. Class 3's lower AUC (0.79) likely reflects significant internal variability within the majority class creating overlap with neighbouring groups.

Overall, the neural network model performed effectively given non-linearity of the data. The Adam optimiser, with its adaptive learning rate mechanism, achieved smoother and faster convergence and generalisation as compared to SGD.

6. Recommendations and Conclusions

This project successfully demonstrated how a neural network can classify abalone age groups based on their physical measurements. Through systematic experimentation and parameter tuning, the model achieved reliable and generalisable performance on a complex, imbalanced dataset. The final model—a two-layer neural network with 15 neurons per layer, trained using the Adam optimiser with a learning rate of 0.1 managed to yield the best balance between accuracy, stability and convergence.

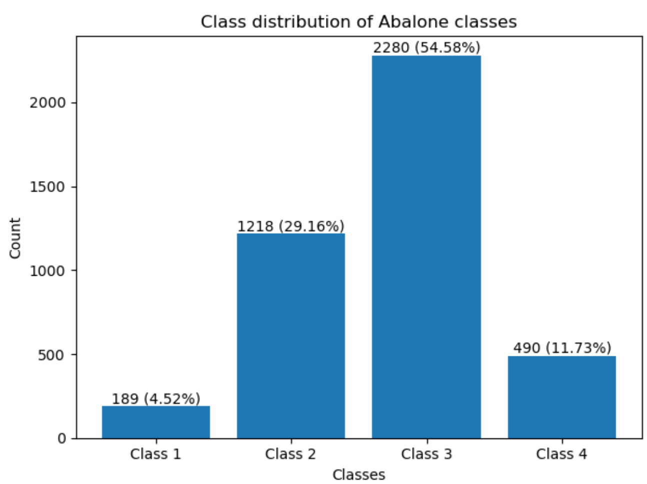
The results highlight several key insights. Firstly, the strong correlations between weight-related features confirm their predictive relevance to abalone age but also suggest potential redundancy that could be mitigated through feature selection. Class imbalances had a significant influence on model accuracy, particularly for the youngest and oldest abalones, which were severely underrepresented. This limitation restricted the network's ability to form robust decision boundaries for these groups.

Recommendations are to implement class-balancing or class-weight adjustments to balance minority class representation. Additionally, exploring alternative model architectures such as deeper networks, data standardisation or performing regularisation methods like dropout could help to further enhance predictive performance.
Nevertheless, the findings confirm that neural networks are capable of effectively modelling non-linear relationships such as the Abalone dataset. With improved data balance and feature refinement, the model's predictive reliability and interpretability could be significantly enhanced, providing a more accurate and efficient alternative to the mundane traditional manual ring-counting methods

7. Appendices

- **Class 1: ring age <= 7**
- **Class 2: 7 < ring age <=10**
- **Class 3: 10 < ring age <= 15**
- **Class 4: ring age > 15**

Appendix 1



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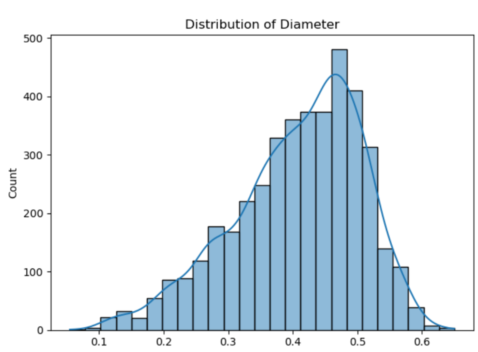

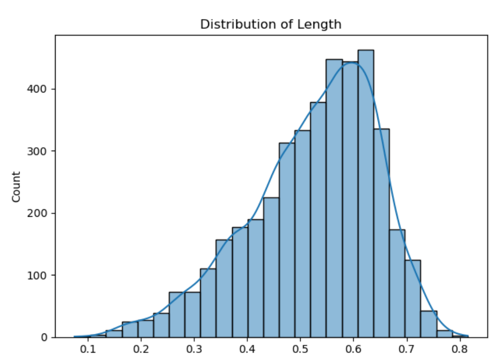

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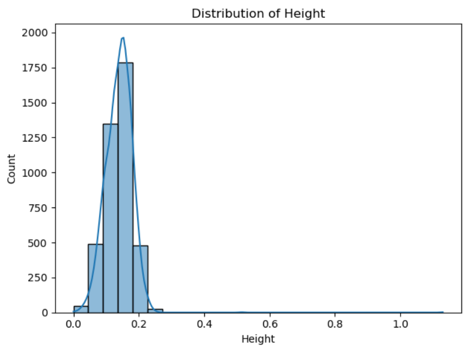

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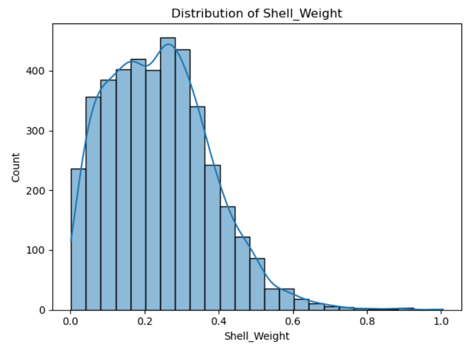

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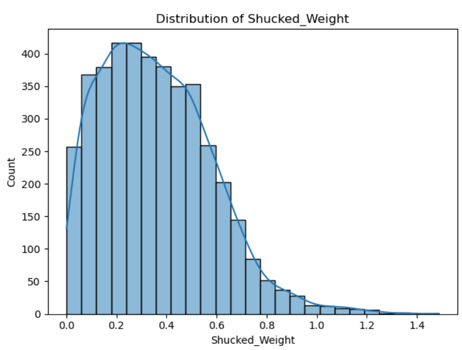

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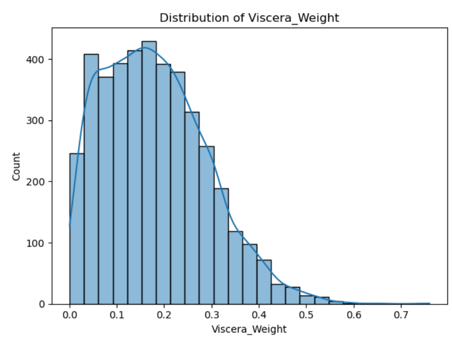

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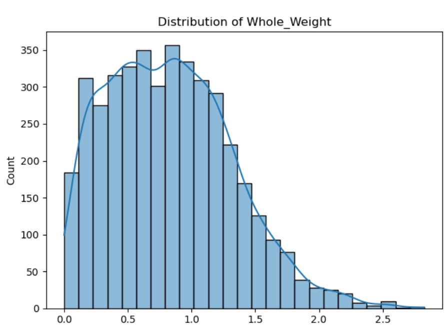

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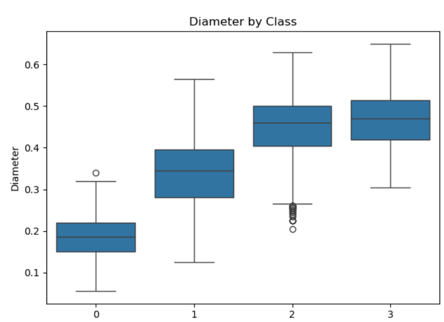

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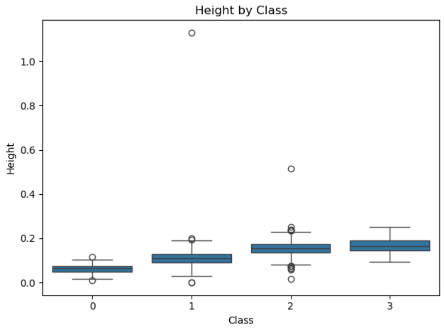
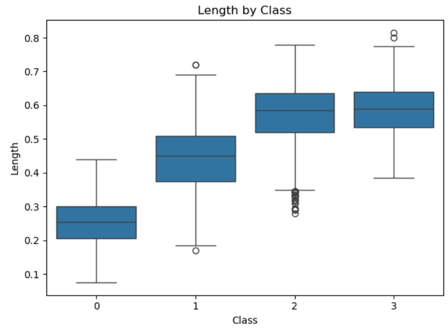
Appendix 10





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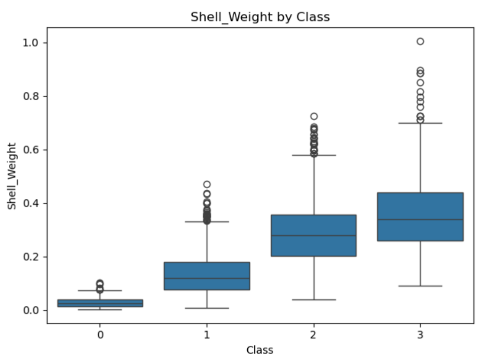
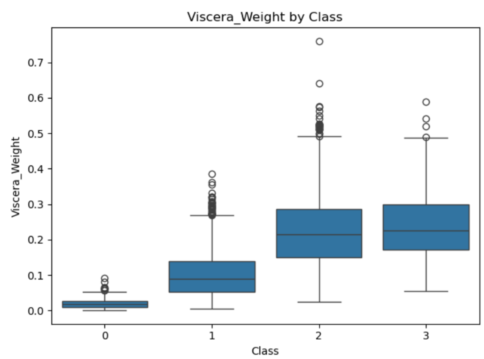
Appendix 12





Appendix 13

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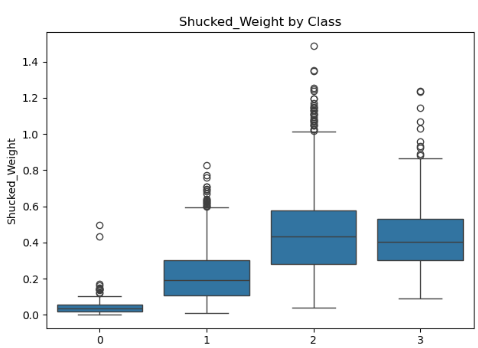

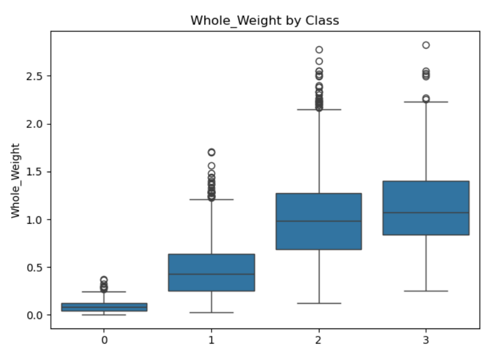

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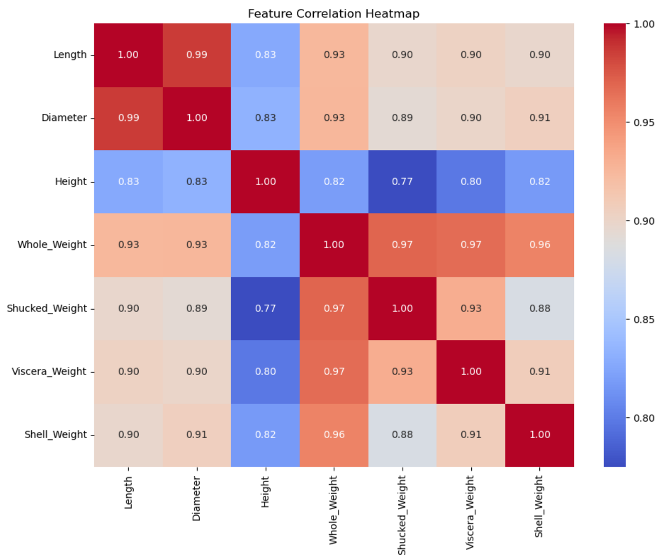
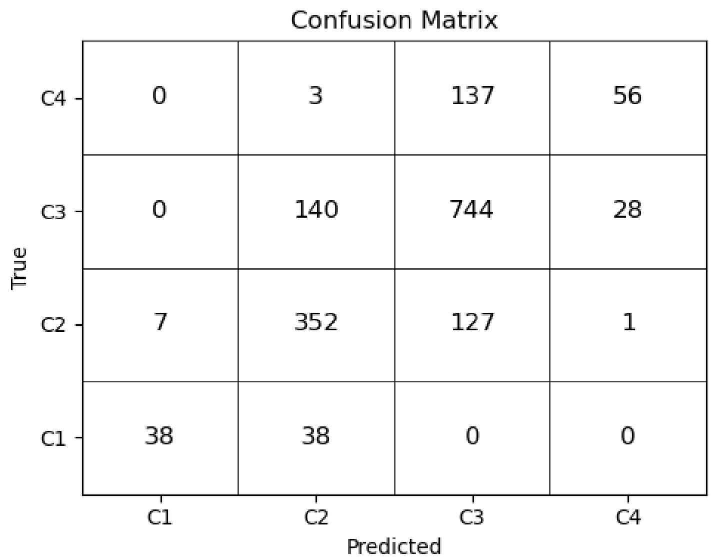
Appendix 16



Appendix 17

	Layers	Hidden Units	Optimizer	Learning Rate	Train Acc	Test Acc
1	1 layer	5	SGD	0.01	0.66 ± 0.00 (± 0.00)	0.67 ± 0.00 (± 0.00)
2	1 layer	10	SGD	0.01	0.66 ± 0.01 (± 0.00)	0.67 ± 0.00 (± 0.00)
3	1 layer	15	SGD	0.01	0.66 ± 0.00 (± 0.00)	0.67 ± 0.00 (± 0.00)
4	1 layer	20	SGD	0.01	0.66 ± 0.00 (± 0.00)	0.67 ± 0.00 (± 0.00)
5	1 layer	15	SGD	0.0001	0.51 ± 0.15 (± 0.10)	0.51 ± 0.14 (± 0.10)
6	1 layer	15	SGD	0.001	0.65 ± 0.02 (± 0.01)	0.65 ± 0.02 (± 0.02)
7	1 layer	15	SGD	0.01	0.66 ± 0.00 (± 0.00)	0.67 ± 0.00 (± 0.00)
8	2 layers	15+15	SGD	0.01	0.68 ± 0.01 (± 0.00)	0.68 ± 0.01 (± 0.00)
9	2 layers	15+15	Adam	0.01	0.73 ± 0.00 (± 0.00)	0.73 ± 0.01 (± 0.00)

Appendix 18



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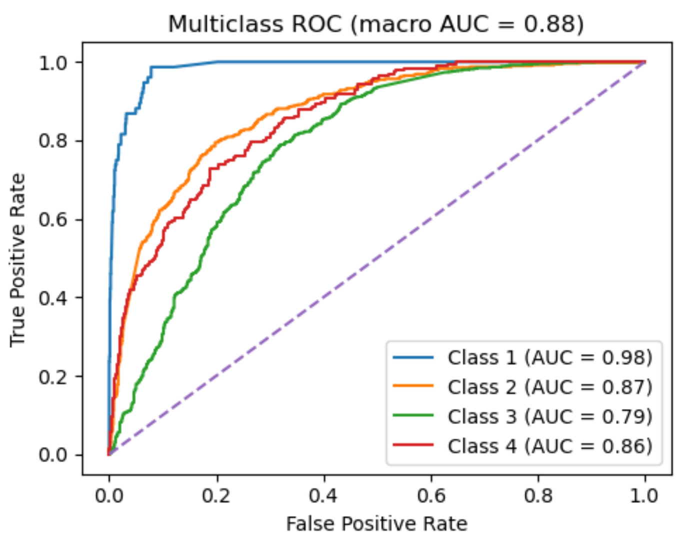

Appendix 20

8. References

"Abalone," *UCI Machine Learning Repository*, University of California, Irvine, CA. [Online]. Available: https://archive.ics.uci.edu/dataset/1/abalone. [Accessed: Oct. 01, 2025].